FUNCTIONAL REQUIREMENTS

FR ID	CATEGORY	PRIORITY	REQUIREMENT DETAILS
FR-01	Catalogue	Must	System shall allow users to browse movies currently showing or coming soon.
FR-02	Catalogue	Must	System shall provide a search bar to find movies by title, genre, or director.
FR-03	Catalogue	Must	System shall display movie details including cast, duration, rating, and synopsis.
FR-04	AI	Should	System shall provide personalized movie recommendations based on user history.
FR-05	Booking	Must	System shall provide an interactive seating map for real-time seat selection.
FR-06	Booking	Must	System shall support multiple ticket types (Adult, Child, Senior, Student).
FR-07	Booking	Should	System shall allow users to pre-order snacks and beverages during booking.
FR-08	Payment	Must	System shall process secure payments via Integrated Credit/Debit Card API.
FR-09	Payment	Should	System shall support payment via digital wallets (Apple Pay, Google Pay).
FR-10	Notification	Must	System shall send an automated booking confirmation via Email/SMS.
FR-11	Booking Mgmt	Must	System shall allow users to view their current and past booking history.
FR-12	Booking Mgmt	Must	System shall generate a unique QR code for each valid ticket purchased.
FR-13	Booking Mgmt	Should	System shall allow booking cancellation based on defined cinema refund policies.
FR-14	Profile	Must	System shall allow users to create and manage personal accounts and passwords.
FR-15	Profile	Should	System shall store user preferences (e.g., favorite genres, preferred cinema locations).
FR-16	Validation	Must	System shall provide a staff interface to scan and validate QR codes at entry.
FR-17	Admin	Must	System shall allow administrators to add, edit, or remove movies from the catalogue.
FR-18	Admin	Must	System shall allow administrators to set showtimes and assign movies to halls.
FR-19	Admin	Must	System shall provide tools for managing user accounts and roles (Staff vs Admin).
FR-20	Admin	Should	System shall generate daily sales and attendance reports for management.

2. Non-Functional Requirements (NFR)

NFR ID	TYPE	PRIORITY	REQUIREMENT DETAILS
NFR-01	Performance	MUST	System shall respond to search queries in less than 2.0 seconds under peak load.
NFR-02	Security	MUST	User passwords shall be hashed using BCrypt and data encrypted via SSL/TLS.
NFR-03	Usability	SHOULD	System interface shall be fully responsive on mobile, tablet, and desktop devices.
NFR-04	Reliability	MUST	System shall maintain an uptime of 99.9% excluding scheduled maintenance.
NFR-05	Maintainability	SHOULD	Backend architecture shall use modular microservices for ease of component updates.
NFR-06	Compatibility	MUST	System web portal shall be compatible with Chrome, Safari, and Edge browsers.
NFR-07	Security	MUST	The payment module shall be PCI-DSS compliant for all financial transactions.
NFR-08	Performance	MUST	System shall support up to 5,000 concurrent active users during peak show releases.

Activate Wir
Go to Settings to

REQUIREMENTS TRACEABILITY MATRIX

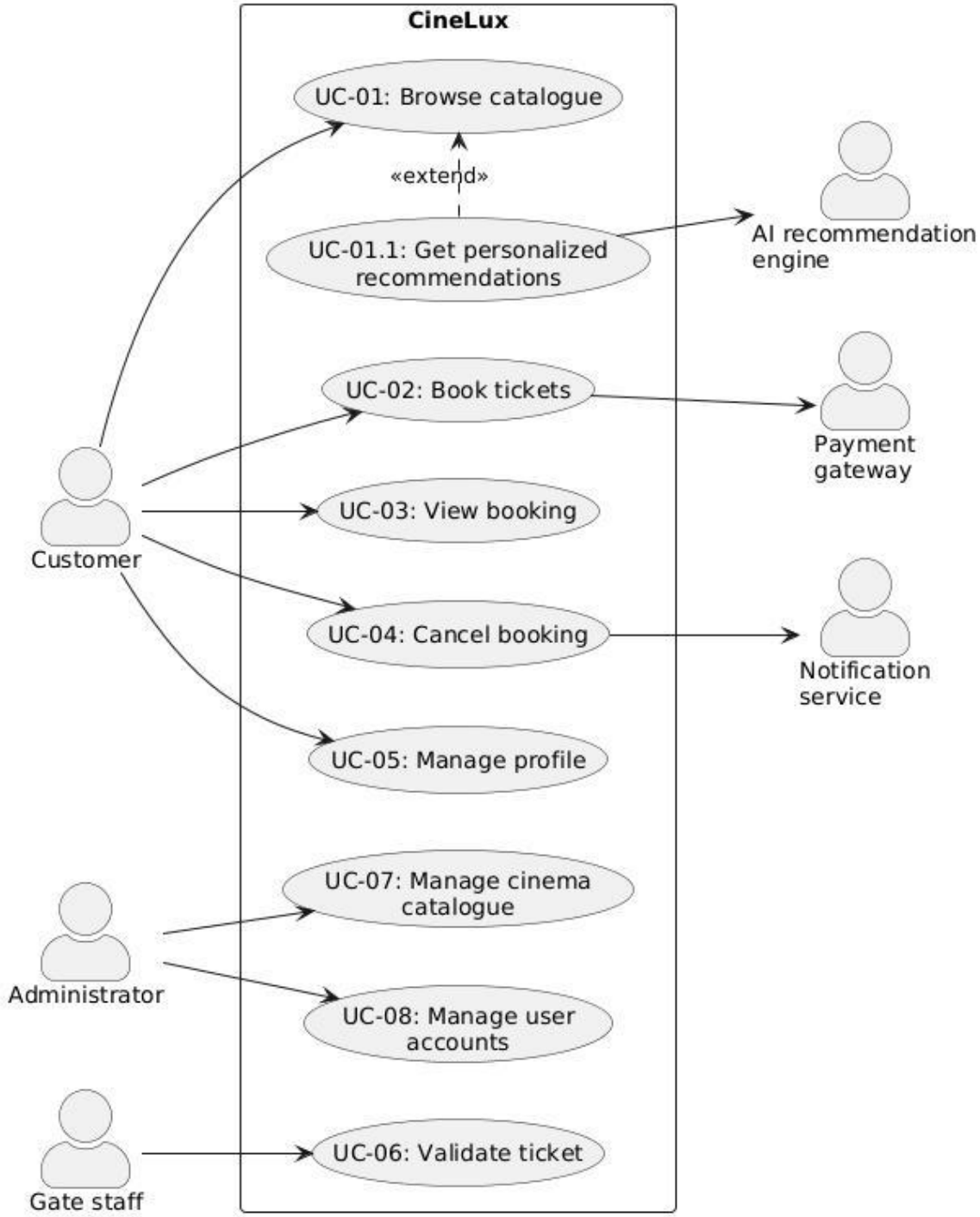
Traceability: FR → UC → TC

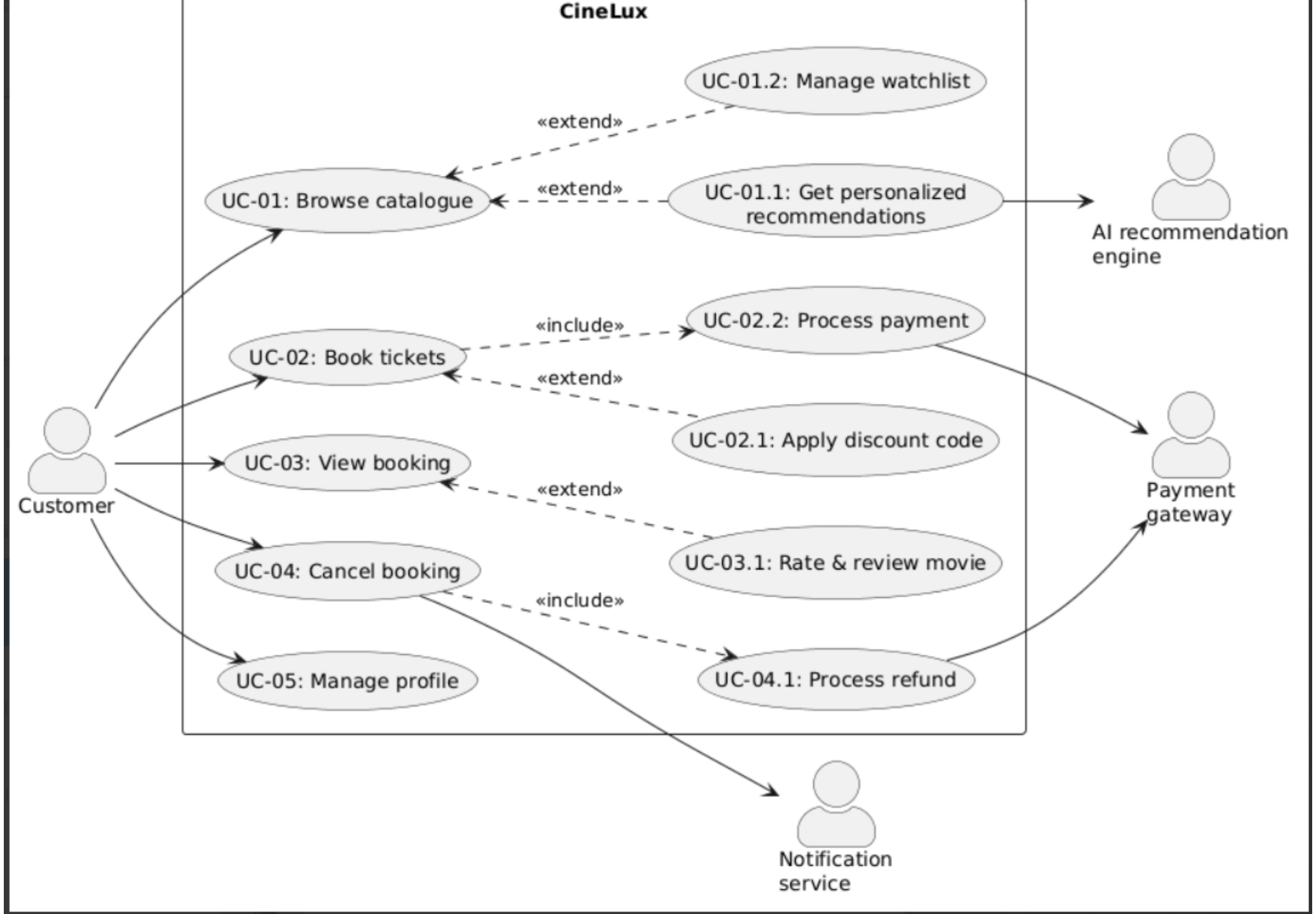
REQ ID	REQUIREMENT TITLE	UC	ANALYSIS	SYS DES	DET DES	CODING	UI	TEST CASE (TC ID + DESCRIPTION)
FR-01	Catalogue Browsing	UC-01						TC-01: Verify landing page movie display
FR-02	Search Function	UC-01						TC-02: Verify search results for valid title
FR-03	Movie Details	UC-01						TC-03: Verify detail page content accuracy
FR-04	AI Recommendations	UC-01.1						TC-04: Verify personalization logic output
FR-05	Seat Selection	UC-02						TC-05: Verify seat lock upon selection
FR-06	Ticket Tiering	UC-02						TC-06: Verify pricing for Child vs Adult
FR-07	Snack Upselling	UC-02						TC-07: Verify snack addition to cart
FR-08	Gateway Payment	UC-02						TC-08: Verify successful card transaction
FR-09	Digital Wallet	UC-02						TC-09: Verify Apple Pay integration
FR-10	Auto Confirmations	UC-02						TC-10: Verify email delivery post-payment
FR-11	History Tracking	UC-03						TC-11: Verify view past tickets list
FR-12	QR Generation	UC-03						TC-12: Verify QR code image rendering
FR-13	Cancellation Policy	UC-04						TC-13: Verify refund logic < 24h notice
FR-14	User Account Mgmt	UC-05						TC-14: Verify password reset functionality
FR-15	Preferences Storage	UC-05						TC-15: Verify genre preference saving
FR-16	Ticket Validation	UC-06						TC-16: Verify QR scan status "Validated"
FR-17	Movie Inventory	UC-07						TC-17: Verify admin "Add Movie" update
FR-18	Showtime Scheduling	UC-07						TC-18: Verify conflict check in schedule
FR-19	Role Permissions	UC-08						TC-19: Verify Staff restricted access
FR-20	Analytics Report	UC-08						TC-20: Verify PDF export of sales data

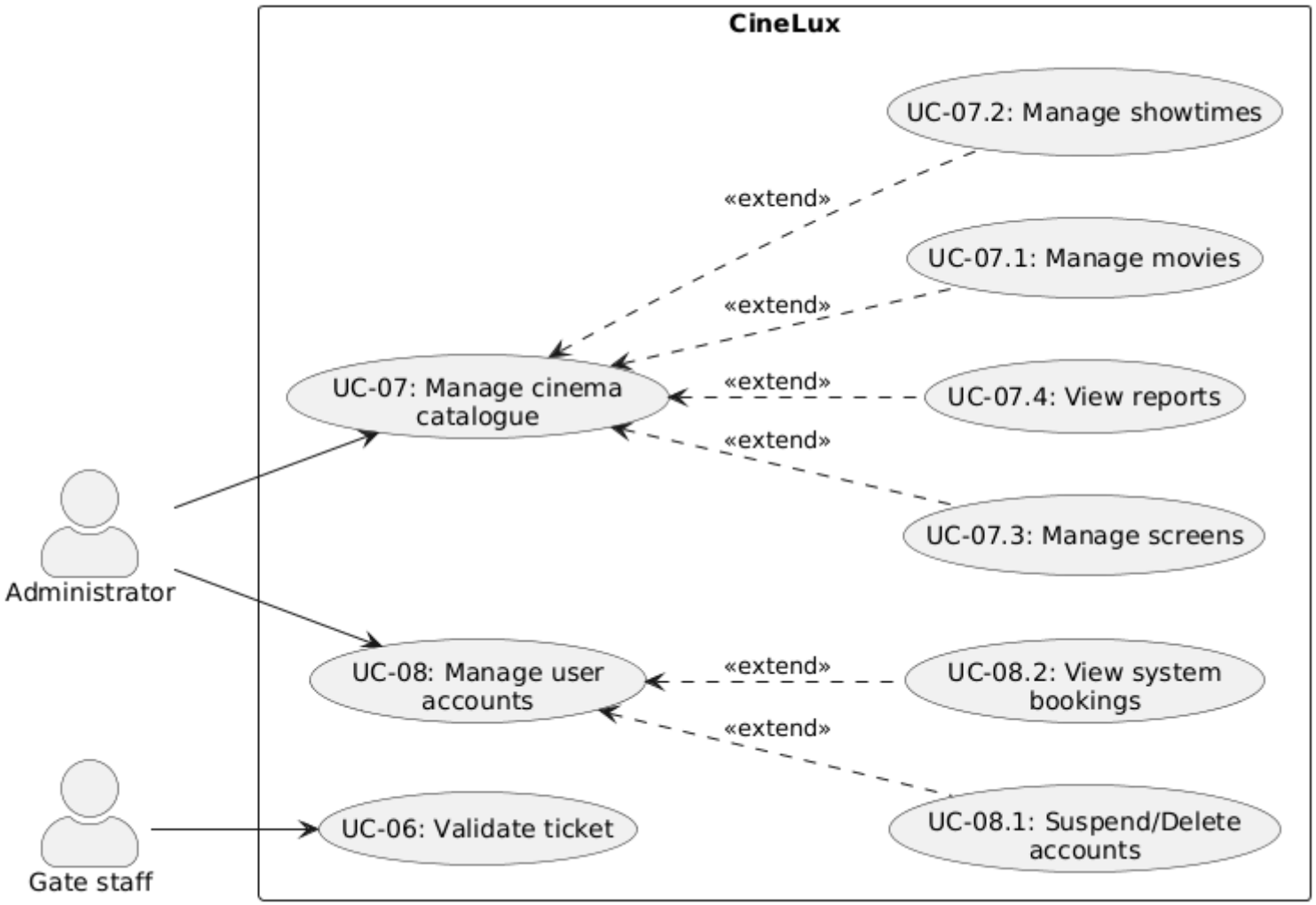
SYSTEM TEST CASES

Cinema Booking System | TC-01 to TC-21

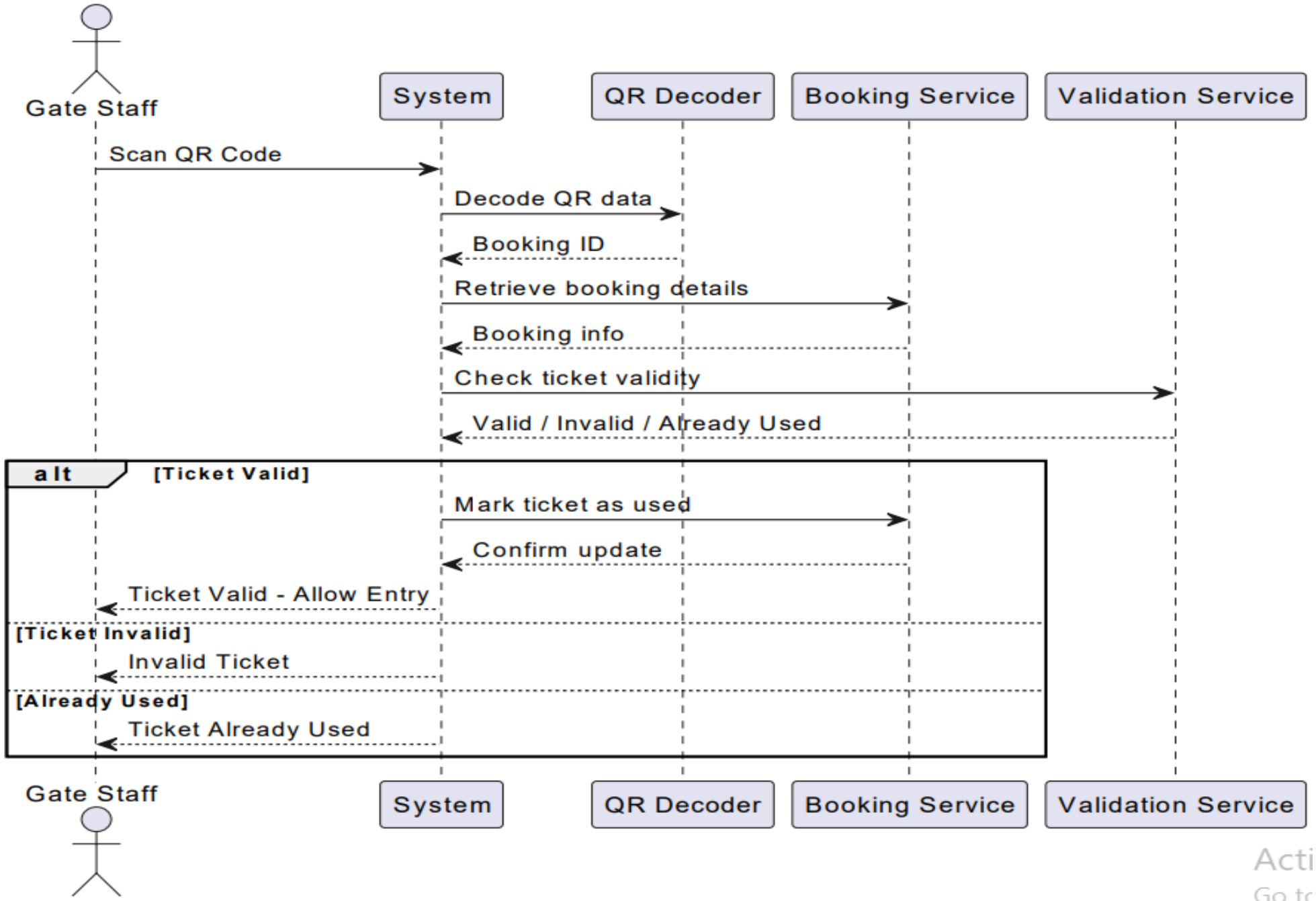
TC ID	TITLE	TEST DATA	STEPS	EXPECTED RESULT
TC-01	Valid Browse	Null	1. Open URL. 2. Observe list.	Landing page displays current movies.
TC-02	Search Positive	"Avatar"	1. Enter title in search box. 2. Click Search.	"Avatar" details are listed.
TC-03	Search Negative	"@#$%"	1. Enter special chars. 2. Click Search.	Display "No results found" message.
TC-04	Recommendation	User ID: 101	1. Log in. 2. View "For You" section.	Personalized list appears based on past viewings.
TC-05	Seat Select Pos	Hall A, C10	1. Select movie. 2. Select Seat C10.	Seat turns Blue (Selected).
TC-06	Seat Select Neg	Hall A, C10	1. Select already booked seat.	Seat is disabled/Red; selection blocked.
TC-07	Payment Success	Valid Visa Card	1. Enter card details. 2. Authorize.	Payment accepted; confirmation screen shown.
TC-08	Payment Decline	Exp. Card	1. Enter expired card. 2. Authorize.	Error: "Card Expired"; transaction halted.
TC-09	Confirm Email	user@mail.com	1. Complete booking. 2. Check Inbox.	Confirmation email received with PDF ticket.
TC-10	View Booking	Valid Login	1. Click My Bookings.	List shows all future and past shows.
TC-11	Cancel Booking	Ticket #404	1. Select active ticket. 2. Click Cancel.	Ticket status updated to "Refunded".
TC-12	Profile Update	New Ph: 0111	1. Edit profile. 2. Save new phone.	Success message; DB shows updated phone.
TC-13	Staff Validation	QR Code #99	1. Scan valid QR. 2. Verify screen.	Screen shows "Verified - Admit 1".
TC-14	Invalid QR Scan	Fake QR	1. Scan random code.	Screen shows "Invalid Code - Entry Denied".
TC-15	Used QR Scan	QR Code #99	1. Scan same QR twice.	Screen shows "Error: Ticket Already Used".
TC-16	Admin Add Movie	"Inception"	1. Add new movie in Admin Panel. 2. Save.	"Inception" visible on customer landing page.
TC-17	Conflict Sched	Hall A, 2PM	1. Book Hall A for two movies at 2PM.	System throws "Schedule Overlap Error".
TC-18	Admin Suspend	User ID: 505	1. Suspend account. 2. Try Login as user.	Login fails: "Account Suspended".
TC-19	Empty Search	""	1. Submit empty search field.	Show all movies or prompt for input.
TC-20	Snack Order	Popcorn XL	1. Add Popcorn to ticket. 2. Checkout.	Invoice totals movie price + snack price.

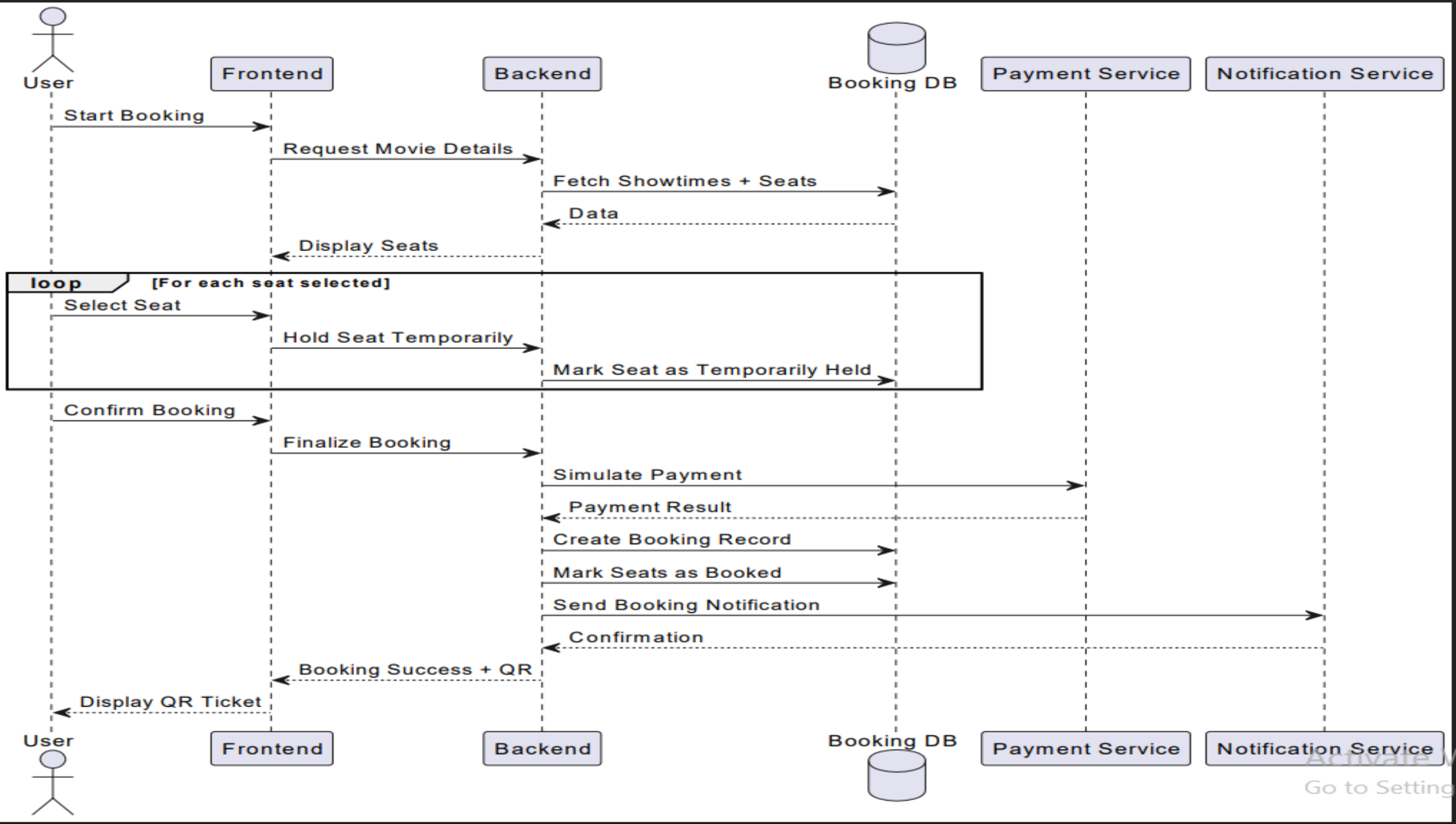

CineLux
UC-01: Browse catalogue
«extend»
UC-01.1: Get personalized recommendations
AI recommendation engine
UC-02: Book tickets
Payment gateway
Customer
UC-03: View booking
UC-04: Cancel booking
Notification service
UC-05: Manage profile
UC-07: Manage cinema catalogue
Administrator
UC-08: Manage user accounts
Gate staff
UC-06: Validate ticket

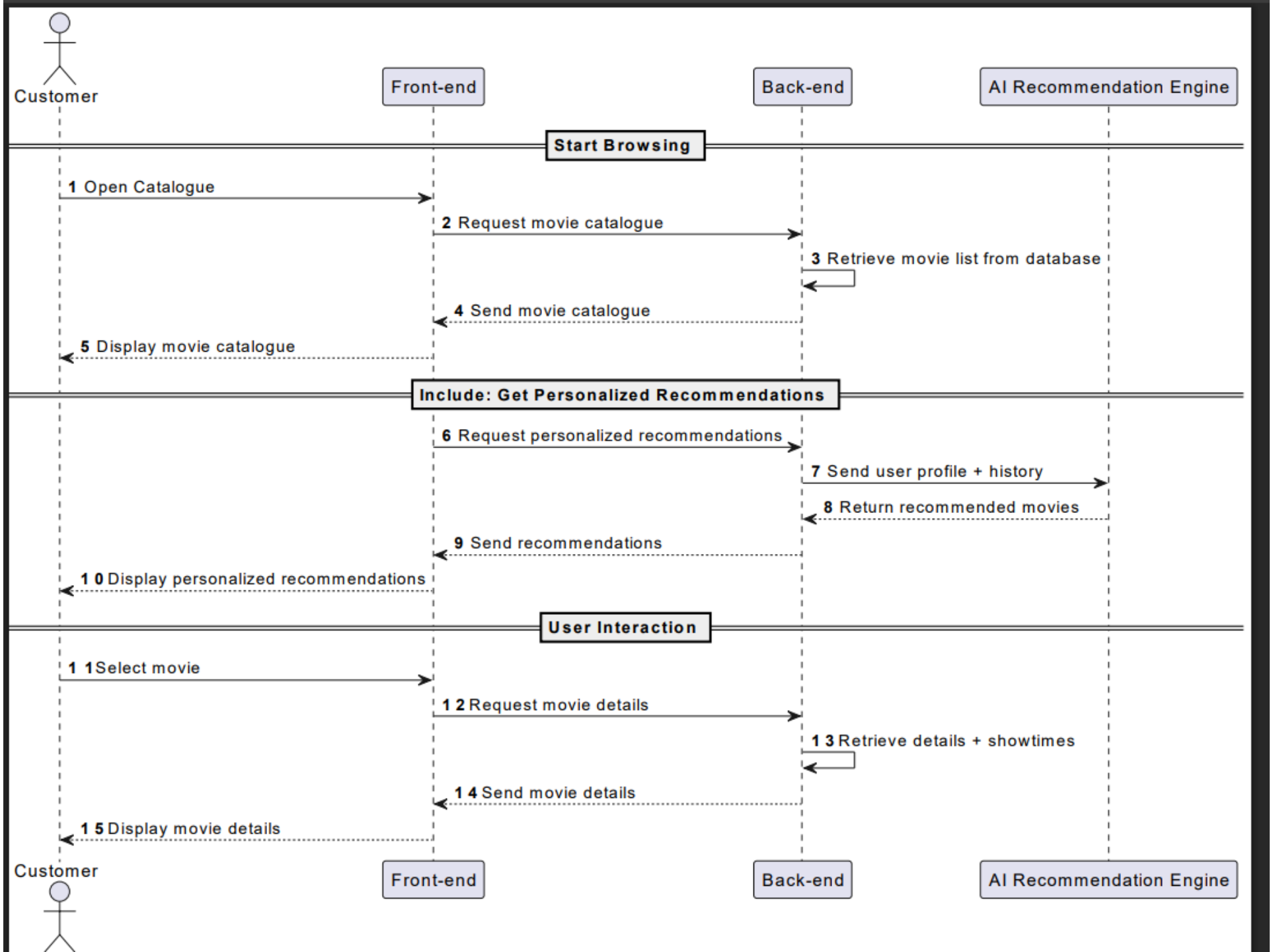

CineLux
UC-01.2: Manage watchlist
«extend»
UC-01: Browse catalogue
«extend»
UC-01.1: Get personalized recommendations
AI recommendation engine
«include»
UC-02.2: Process payment
UC-02: Book tickets
«extend»
UC-02.1: Apply discount code
Customer
UC-03: View booking
«extend»
Payment gateway
UC-04: Cancel booking
UC-03.1: Rate & review movie
«include»
UC-05: Manage profile
UC-04.1: Process refund
Notification service

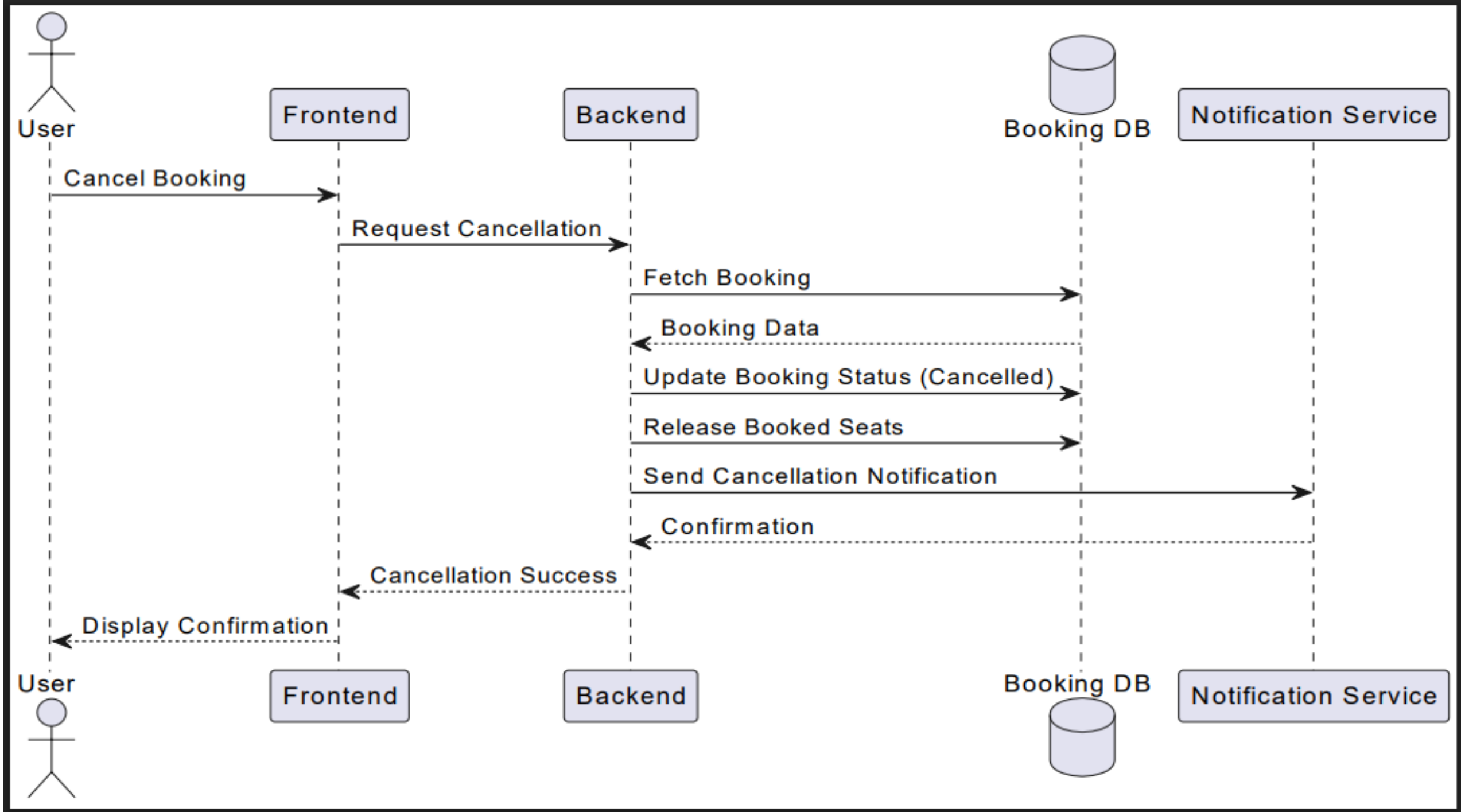

CineLux
UC-07.2: Manage showtimes
«extend»
UC-07.1: Manage movies
«extend»
UC-07: Manage cinema catalogue
«extend»
UC-07.4: View reports
«extend»
UC-07.3: Manage screens
Administrator
UC-08: Manage user accounts
«extend»
UC-08.2: View system bookings
«extend»
UC-08.1: Suspend/Delete accounts
Gate staff
UC-06: Validate ticket

Sequence Diagram - UC-06 Validate Ticket




User
Frontend
Backend
Booking DB
Payment Service
Notification Service
Start Booking
Request Movie Details
Fetch Showtimes + Seats
Data
Display Seats
loop
[For each seat selected]
Select Seat
Hold Seat Temporarily
Mark Seat as Temporarily Held
Confirm Booking
Finalize Booking
Simulate Payment
Payment Result
Create Booking Record
Mark Seats as Booked
Send Booking Notification
Confirmation
Booking Success + QR
Display QR Ticket
User
Frontend
Backend
Booking DB
Payment Service
Notification Service


Customer
Front-end
Back-end
AI Recommendation Engine
Start Browsing
1 Open Catalogue
2 Request movie catalogue
3 Retrieve movie list from database
4 Send movie catalogue
5 Display movie catalogue
Include: Get Personalized Recommendations
6 Request personalized recommendations
7 Send user profile + history
8 Return recommended movies
9 Send recommendations
1 0 Display personalized recommendations
User Interaction
1 1 Select movie
1 2 Request movie details
1 3 Retrieve details + showtimes
1 4 Send movie details
1 5 Display movie details
Customer
Front-end
Back-end
AI Recommendation Engine


User
Frontend
Backend
Booking DB
Notification Service
Cancel Booking
Request Cancellation
Fetch Booking
Booking Data
Update Booking Status (Cancelled)
Release Booked Seats
Send Cancellation Notification
Confirmation
Cancellation Success
Display Confirmation
User
Frontend
Backend
Booking DB
Notification Service